THE VALUATIVE CRITERION FOR ALGEBRAIC SPACES IS PRESERVED BY COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The existence case is presented through the displayed two-step diagram and then spelled out in words; both are inside the counted proof span.

Editorial note. The counted spans carry no references at all, so the card needs no navigation stubs beyond the definition pointer implied by the statement shape.

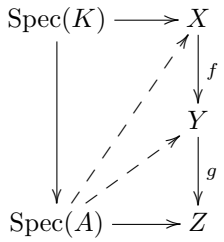
Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The shape of the valuative criterion diagram. Definition definition-valuative-criterion, used at proof lines 8738-8745. Fixes the solid diagram over the composition Z <- Y <- X.
- Uniqueness and existence parts of the criterion for the two factors. used at proof lines 8746-8750 and 8765-8768. Are applied successively to build and compare the dotted arrows.
- Valuation rings under field extension and domination are again of the same kind. used at proof lines 8754-8762. Allows the second application of the criterion to use the field extension produced by the first.

Lemma 0.2. *The composition of two morphisms of algebraic spaces which satisfy the (existence part of, resp. uniqueness part of) the valuative criterion satisfies the (existence part of, resp. uniqueness part of) the valuative criterion.*

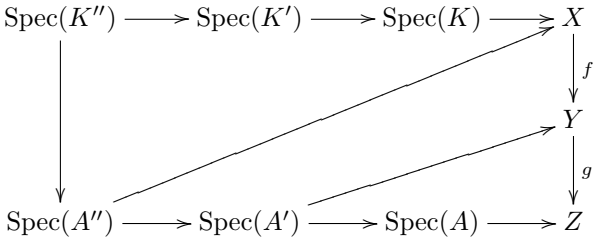
Proof. Let $f : X \to Y$, $g : Y \to Z$ be morphisms of algebraic spaces over the scheme S. Consider a solid commutative diagram of the following shape



If we have the uniqueness part for g, then there exists at most one north-west dotted arrow making the diagram commute. If we also have the uniqueness part for f, then we have at most one north-north-west dotted arrow making the diagram

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

commute. The proof in the existence case comes from contemplating the following diagram



Namely, the existence part for g gives us the extension K', the valuation ring A' and the arrow $\mathrm{Spec}(A') \to Y$, whereupon the existence part for f gives us the extension K'', the valuation ring A'' and the arrow $\mathrm{Spec}(A'') \to X$. $\square$